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Influence of Hydrogen Bonding between Ions of Like Charge on the Ionic Liquid Interfacial Structure at a Mica Surface

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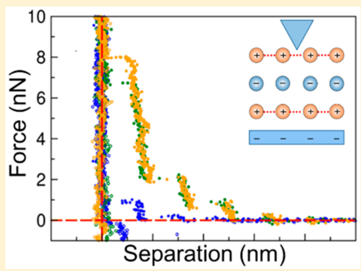
Supporting Information

ABSTRACT: Ionic liquids (ILs) have attracted increasing interest in science and technology because of their remarkable properties, which can be tuned via varying ion structures to control the relative strengths of Coulomb interactions, hydrogen bonding (H-bonding), and dispersion forces. Here we use atomic force microscopy to probe the interfacial nanostructures of hydroxy functionalized ILs at negatively charged mica surfaces. H-bonding between hydroxy functionalized cations (c-c) produces cation clusters and a stronger interfacial nanostructure. H-bond stabilized cation clusters form despite opposing electrostatic repulsions between charge groups, cation−anion (c-a) electrostatic attractions, and (c-a) H-bonds. Comparison of ILs with and without OH functionalized cations shows directional H-bonding enhances interfacial structure more strongly than the dispersion forces between alkyl groups. These findings reveal a new means of controlling IL interfacial nanostructure via H-bonding between like-charged ions, which impact diverse areas including electrochemical charge storage (batteries and catalysis), electrodeposition, lubrication, etc.



In recent years, ionic liquids (ILs) have attracted increasing interest in science and technology owing to their unique properties, which often include a wide electrochemical window,[1,2] very low vapor pressure,[3,4] and high thermal stability.[5] Per definition, a salt can be called an IL if it melts at low temperature and a room temperature ionic liquid (RTIL) if the melting point is below 25 °C. As ionic liquids consist entirely of ions, Coulomb forces are ubiquitous, but properties are governed by a subtle balance of electrostatics, H-bonding, and dispersion forces.[6] In particular, local and directional H-bonding has significant influence on IL behavior.[7]

In most ILs, the H-bonding motif is exclusively the Coulomb-enhanced interaction between a cation and an anion (c-a). However, in hydroxy-modified ILs, H-bonding between ions of like charge, which in the present case involves the positively charged cations (c-c), has recently been shown to be an important driver of bulk liquid structure, leading to H-bonded cation clusters in the bulk liquid phase.[8−11] Fourier transform infrared (FT-IR) measurements clearly showed two distinct hydrogen bonded species, and that (c-c) H-bonds are stronger than (c-a) H-bonds as indicated by the magnitudes of the redshifts of their vibrational bands.[8−11] The formation of these (c-c) clusters is mediated by surrounding counterions and was observed to become more prominent at lower temperature (ca. 213 K).

The diffuse nature of the bulk IL vibrational bands provides only a qualitative picture of the local interaction. To compensate, gas-phase isolated +1 charged clusters were investigated. Chain-like dimers and trimers, where two or three OH groups of the cations form a linear hydrogen-bonded chain from the OH group of one cation to the OH group of the next cation, terminated by an H-bond from the terminal OH group to an anion, were identified.[12,13] Very recently neutron diffraction experiments, supported by MD simulations and DFT calculations, elucidated the bulk nanostructure of the hydroxy functionalized IL [HOC_4Py][NTf_2]. It was found that (c-c) H-bonds are significantly stronger (shorter and straighter) than these for (c-a) and account for ∼20% of the H-bonds present.[14]

The existence of supramolecular cation clusters is expected to have particular impact on the applications of ILs at interfaces and of supported ILs, including heterogeneous catalysis,[15] wood treatment,[16] lubrication,[17] electrochemistry,[18−21] and charge or energy storage.[22] The presence of macroscopic interfaces enforces different ion arrangements compared to the bulk, affecting the structure and dynamics of interfacial processes. Thus, a deep understanding of the factors that control ion arrangements is critical for such applications of ILs.

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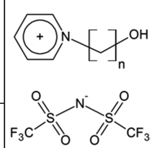
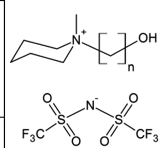
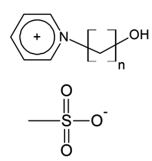
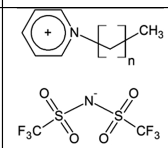
Table 1. Name, Abbreviation, Structure, Molecular Weight (M), Density (ρ), Ion Pair Diameter (D_m), and Measured Separation Layer Thicknesses (Δsep)

IL	Abbreviation	Structure	M [g]	ρ [g·cm^{-3}]	D_m [nm]	$\Delta sep.$ [nm]
1-(2-hydroxyethyl)pyridinium-bis(trifluoromethanesulfonyl)imide	$[HOC_2Py][NTf_2]$		404.31	1.595	0.749	0.770
1-(3-hydroxypropyl)pyridinium-bis(trifluoromethanesulfonyl)imide	$[HOC_3Py][NTf_2]$		418.34	1.557	0.764	0.756
1-(4-hydroxybutyl)pyridinium-bis(trifluoromethanesulfonyl)imide	$[HOC_4Py][NTf_2]$		432.37	1.520	0.779	0.794
1-(2-hydroxyethyl)methylpiperidinium-bis(trifluoromethanesulfonyl)imide	$[HOC_2MPip][NTf_2]$		424.39	1.500	0.777	0.780
1-(3-hydroxypropyl)methylpiperidinium-bis(trifluoromethanesulfonyl)imide	$[HOC_3MPip][NTf_2]$		438.41	1.470	0.791	0.763
1-(4-hydroxybutyl)methylpiperidinium-bis(trifluoromethanesulfonyl)imide	$[HOC_4MPip][NTf_2]$		452.44	1.444	0.804	0.804
1-(3-hydroxypropyl)pyridinium-methanesulfonate	$[HOC_3Py][OMs]$		233.29	1.297	0.668	0.564
1-(4-hydroxybutyl)pyridinium-methanesulfonate	$[HOC_4Py][OMs]$		247.32	1.310	0.679	0.687
1-pentylpyridinium-bis(trifluoromethanesulfonyl)imide	$[C_5Py][NTf_2]$		430.39	1.436	0.792	0.790

Here we use atomic force microscopy (AFM) to examine the interfacial nanostructure of ILs near a mica surface, which is used as a model anionic substrate. We have developed a series of nine OH-functionalized ILs in order to explore how three key structural factors known to influence (c-c) interactions in the bulk liquid affect interfacial IL structure (see Table 1).[10−13] Results are compared with the unfunctionalized IL 1-pentylpyridinium-bis(trifluoromethanesulfonyl)imide ($[C_5Py][NTf_2]$), which has the same shape and molecular volume as $[HOC_4Py][NTf_2]$.

(1) Increasing the chain length of the 1-(x-hydroxyalkyl)pyridinium- bis(trifluoromethanesulfonyl)imide ($[HOC_xPy][NTf_2]$) ILs from ethyl to butyl increases the distance between the OH functional group and the positive charge center of the cation, weakening the like-charge repulsion and favoring the (c-c) H-bond.

(2) Replacing the delocalized charge on the polarizable pyridinium cation with 1-(x-hydroxyalkyl)methylpiperidinium ($[HOC_xMPip][NTf_2]$), which possesses a localized charge on the nitrogen, less effectively compensates for the negative charge of the anion and decreases (c-c) H-bonds.

(3) Replacing the $[NTf_2]^-$ anion with methanesulfonate ($[OMs]^-$), which coordinates strongly to the OH group, leads exclusively to (c-a) H-bonding.

ILs were synthesized at Universität Rostock. Synthesis details are provided in the Supporting Information. The ILs were dried under vacuum to a water content below 10 ppm, determined by Karl Fischer titration. To obtain the force curves in the AFM experiments, we used a Nanoscope IV multimode AFM (Veeco Instruments) in contact mode with a sharpened Si tip. The spring constant was 0.06 N/m. The ILs were held in an AFM fluid cell, sealed by a silicone O-ring. These were cleaned by rinsing several times with Milli-Q water,

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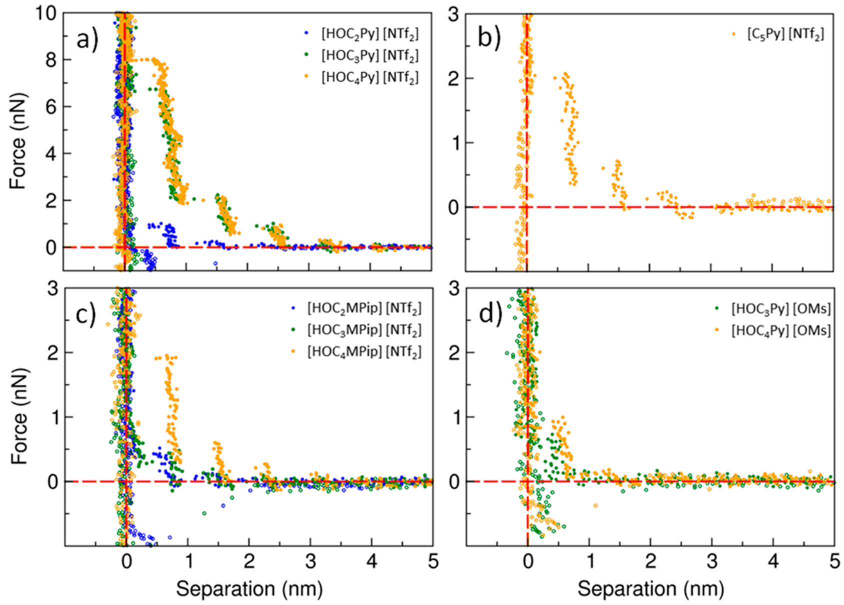


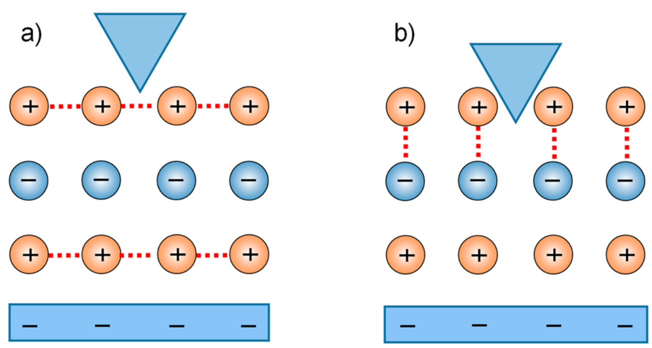
Figure 1. Force separation profiles for the subsets: (a) [HOC$_n$Py][NTf$_2$], (b) [C$_5$Py][NTf$_2$], (c) [HOC$_n$MPip][NTf$_2$], and (d) [HOC$_n$MPip][OMs]. All profiles show the approach (filled circles) of the AFM tip to the mica surface as well as the retraction (open circles).

followed by rinsing several times with ethanol, and then dried using filtered nitrogen. The cantilever was carefully rinsed with ethanol and irradiated with ultraviolet light for 10 min prior to every experiment to avoid contaminations. A fresh mica surface was obtained for each experiment by removing the top layer of a mica wafer with an adhesive tape. The mica surface thus obtained is clean and atomically smooth. After the AFM cell was filled with IL, the system was equilibrated for at least 1 h. The scan rate was 0.2 Hz and the scan size was 50 nm, resulting in a tip velocity of 20 nm/s. All experiments were performed at room temperature and repeated at least three times. The experimental conditions allow the investigation of only liquids at room temperature, which makes it impossible for us to study [HOC$_2$Py][OMs], [C$_3$Py][NTf$_2$], and [C$_4$Py][NTf$_2$], which are solid at room temperature.

Figure 1a shows the effect of alkyl chain length on the force separation profiles (FSPs) for the [HOC$_x$Py][NTf$_2$] series. During the approach of the tip to the surface, the force acting on the tip remains constant at separations greater than 4 nm. This force is set to zero and indicates bulk structure. At smaller separations the force on the tip increases abruptly; the tip jumps then inward as it encounters a series of regularly spaced, repulsive barriers. These periodic force repulsions indicate the bulk structure has transitioned into a more ordered layered structure near the negatively charged mica surface (see Scheme 1). When the tip encounters the outermost detectable interfacial layer, it pushes against it, leading to a repulsive force. Once the force is sufficiently high, it breaks through the layer and "jumps" to the next one, whereupon the force increases again and the process repeats. The closer the tip is to the surface, the more structured the interfacial layers and the higher the breakthrough force. At zero separation, our AFM imaging[23] and force curve investigations[24,25] revealed a strongly adsorbed cation layer that resists "squeeze out" even at high force because of strong Coulombic attractions between cations and the negatively charged mica surface.

The jump distance is the interfacial layer thickness (Δsep). In previous studies, Δsep has been shown to be well-approximated by the ion pair diameter, D_m, which is calculated simply from the macroscopic density and the molecular weight, assuming a cubic packing geometry.[26−28] The agreement here is also quite good (c.f. Table 1).

Scheme 1. Simplified Illustration of Possible Local Structures at a Negatively Charged Interface[a]



[a]The cation-enriched layers are shown as entirely cationic, and the anion-enriched layers are depicted as entirely anionic for the sake of simplicity. Panel (a) illustrates the stabilizing effect of enhanced H-bonding between the cations on the interfacial layer formation. The tip pushes against the layer during the approach to the surface. Panel (b) illustrates the exclusive occurrence of anion−cation H-bonds and the absence of cation−cation H-bonds. The ion pairs thus obtained can easily move together and be displaced by the tip without facing a force significantly higher than in the bulk phase.

The number of detectable layers increases with chain length of the [HOC$_x$Py][NTf$_2$] ILs in Figure 1a, from three for [HOC$_2$Py][NTf$_2$] to five for [HOC$_4$Py][NTf$_2$]. The break-through forces increase markedly from [HOC$_2$Py][NTf$_2$] to [HOC$_3$Py][NTf$_2$], then more modestly to [HOC$_4$Py][NTf$_2$]. This is because the stabilizing effect of enhanced H-bonding between the cations on the interfacial layers as the separation between OH and cation charge centers increases, as illustrated in Scheme 1a. These findings are in good agreement with the studies of the bulk phase of OH-functionalized cationic clusters. The nonvertical repulsions indicate layer compressi-

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bility due to the flexibility in the underlying structure, which deforms before it finally ruptures.

The importance of (c-c) H-bonds is immediately clear from comparing forces in $[HOC_4Py][NTf_2]$ with the unfunctionalized IL $[C_5Py][NTf_2]$ (see Figure 1b). $[C_5Py][NTf_2]$ and $[HOC_4Py][NTf_2]$ have similar molecular volumes and ion pair diameters, D_m (0.792 and 0.788 nm, respectively). The only difference is that the OH functional group in $[HOC_4Py]^+$ is replaced by a CH_3 group in $[C_5Py]^+$ cation which prevents it from forming (c-c) H-bonds. In the force curves it can be seen that the interfacial layer thicknesses (Δsep = 0.79 nm) are also identical, but the breakthrough forces are much higher in the functionalized IL (8.0 nN vs 2.1 nN). The interfacial layers of $[HOC_4Py][NTf_2]$ are stabilized by H-bonding between the cations, whereas the interfacial layers of the IL $[C_5Py][NTf_2]$ can be stabilized only by weaker, nondirectional van der Waals interactions between the unfunctionalized cation alkyl chains.

Figure 1c shows the effect of reducing cation charge delocalization (polarizability) on the layer integrity for three $[HOC_xMPip][NTf_2]$ ILs. Although periodic repulsions are again observed, the breakthrough forces are lower than those of the corresponding $[HOC_xPy][NTf_2]$ ILs by a factor of 4−10. The strongly localized charge on the nitrogen of the saturated *N*-methylpiperidinium ring of the $[HOC_xMPip]^+$ cations is less able to compensate for the weakly coordinating $[NTf_2]^-$ anions than is the delocalized charge on the $[HOC_xPy]^+$ cations.[10] This leads to a significant decrease of (c-c) H-bonding caused by increased Coulomb repulsion, and the layers weaken to such an extent that $[HOC_4MPip]^+$ is comparable to the non-H-bonding $[C_5Py]^+$.

Careful comparison of measured Δsep to the calculated D_m in these series reveals some systematic deviations (Figure 2).

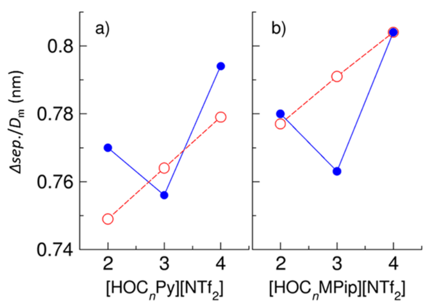

Figure 2. Interfacial layer thickness (Δsep) (blue) and the ion pair diameter, D_m, calculated from the macroscopic density and the molecular weight (red) dependent on the number of carbons in the hydroxyalkyl chain for the ILs: (a) $[HOC_nPy][NTf_2]$ and (b) $[HOC_nMPip][NTf_2]$.

These deviations most likely result from the assumption of isotropic packing for D_m calculation, whereas the ion pair volume of the ILs investigated here is expected to be anisotropic because of the cation chains. We observe a small but significant odd−even alternation in Δsep with increasing cation chain length, which is larger for ILs with even numbers of carbons in the alkyl chain and smaller for ILs with an odd number. Our results indicate that, on average, the anisotropic ion pairs tend to orient their long axes (d_{max}) normal to the surface in the even cases and parallel in the odd cases (see Scheme S1). Odd−even effects are known for many ILs and molecular salts, and their special structural features influence properties like glass transition temperatures, viscosities, and electrical conductivities[29−32] but have not previously been observed at IL interfaces. To study this effect in greater detail, ILs with even longer chain length would be necessary. The interfacial layer thickness (Δsep) is also systematically slightly higher for the piperidinium-based IL, which is also reflected in the calculated ion pair diameters (D_m).

Replacing the weakly coordinating $[NTf_2]^-$ anion with $[OMs]^-$ while keeping the $[HOC_xPy]^+$ cation virtually eliminates all near-surface order in the ILs. The force−separation profiles of $[HOC_3Py][OMs]$ and $[HOC_4Py][OMs]$ show only a single, weak layer adjacent to the mica surface (Figure 1d). From bulk phase IR studies it is known that the $[OMs]^-$ anion is strongly coordinated to the OH functional group of the cations, resulting in strong, Coulomb enhanced (c-a) H-bonds, and the almost complete absence of stabilizing (c-c) H-bonds.[11] The resultant ion pairs are much more mobile than the bigger H-bonded cationic clusters, which means that they can be displaced easily by the AFM tip, as illustrated in Scheme 1b. Only the first two cation layers above the surface are bound strongly enough to the mica surface to form detectable interfacial layers. The interfacial layer thickness (Δsep) of the $[HOC_xPy][OMs]$ ILs is significantly smaller than $[HOC_xPy][\ NTf_2]$ ILs. This is primarily due to the much smaller $[OMs]^-$ anion. (Because the sole detectable layer is probably cation-enriched because of the negative mica charge, we cannot infer other details of near-surface ion-pair arrangements.)

An anomalous increase of capacitance with a decrease in the pore size of a carbon-based porous electric double-layer capacitors has been observed by Chmiola et al.[33] and Largeot et al.[34] Kornyshev et al. proposed a "superionic state" to explain this effect.[35,36] In the "superionic state", image forces exponentially screened out the electrostatic interactions of ions in the interior of a pore; the packing of ions of like charge become easier and is mainly limited by steric interactions. Therefore, the capacitance increased with the decreases of the pore size.[35,36] Because the OH functionalized ionic liquids used in this work can increase the packing of ions of like charge, future work will investigate whether they can further enhance this effect.

In summary, we have used AFM measurements to characterize the nanoscale forces at the solid/IL interface. In particular, we have shown that the formation of (c-c) H-bonds can increase the near-surface structuring of ILs and stabilize the layers, and we have reported the effect of the chain length of the hydroxyalkyl group on the formation of ion-enriched layers at a mica surface. Enhanced H-bonding is due to a reduced Coulomb repulsion, which is caused by a larger distance between the OH group and the positive charge, which means interfacial layers are stabilized. However, a localized positive charge on the cation destabilizes the H-bonding between the cations because of enhanced Coulomb repulsion. By changing the weakly coordinating $[NTf_2]^-$ anion to the strongly coordinating $[OMs]^-$ anion, we revealed the effect of the anion on the near-surface structures. The results suggest that strongly coordinating anions form almost entirely cation−anion pairs via (c−a) H-bonds which can be moved easily by the AFM tip, leading to a decrease of the number of interfacial layers. Finally, we note the stabilizing effect of H-bonding between the functionalized cations by comparing the force separation profile of the IL $[HOC_4Py][NTf_2]$, which shows the most structuring and the highest breakthrough forces, with the force separation profile of the unfunctionalized IL

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$[C_5Py][NTf_2]$, in which the ions are almost identical in size and shape. In conclusion, an ionic liquid's interfacial nanostructure can be controlled by H-bonding between ions of like-charge, the cation polarizability, the anion interaction strength, and hydroxy-alkyl chain length.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpclett.9b03007.

Simplified illustration of possible d_{max}; synthesis of the specific compounds (PDF)

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Notes

The authors declare no competing financial interest.

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